

Exam Date & Time: 14-May-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Examination - April / May -2025

Introduction to Internet of Things [NPT-3202]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) A family installs an IoT-based smart home system that includes smart lighting, security cameras, voice-controlled assistants, and automated climate control. The system collects and processes real-time data to enhance convenience and security. Explain the main components involved in development of such IoT environment? Provide examples. (5)
 - A)
 - B)
- 2) Develop an Arduino Program to read input from a push button and turn on an LED when pressed. (5)
 - A)
 - B)
- 3) Write a sketch that reads temperature data from a DHT11 sensor and displays it on the Serial Monitor (5)
 - A)
 - B)
- 4) Identify the differences between short-range and long-range IoT communication protocols. (5)
 - A)
 - B)
- 5) A large manufacturing company is implementing an Industrial IoT (IIoT) system to automate its smart factory. The factory has thousands of IoT devices, including robotic arms, conveyor belt sensors, temperature monitors, and quality control cameras. These devices need to communicate efficiently with each other and with a central control system. Explain how AMQP is used in establishing this environment? (5)
 - A)
 - B)
- 6) How does a Sensor Cloud integrate with IoT systems? (5)
 - A)
 - B)
- 7) A city government is implementing a smart traffic management system using IoT sensors and cameras to reduce congestion. However, sending all data to the cloud introduces delays. How can fog computing help improve real-time decision-making in (5)
 - A)
 - B)

this scenario?

- 5) Give some real-world examples of SDN implementation.

(5)

A)

B)

Nowadays due to increasing demand, the network infrastructure is becoming complex and inefficient, with traditional networking struggling to adapt to dynamic traffic loads. In such scenario, what are the key factors to consider when choosing an IoT data storage solution? (5)

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